1. Introduction: DTA

What is Drug-Target Affinity?

Continuous value reflecting binding strength between drug and protein. Critical for drug discovery and drug repositioning.

Applications

Drug discovery
Drug repositioning
Lead optimization

Why Computational?

Need: Quantitative predictions vs. binary classification
Big data + deep learning

2. Motivation & Inspiration

Drug Development Crisis

$2.8B	10-15y	Most fail
Cost	Timeline	Approval

Why Deep Learning?

📊 Quantitative Predictions

Need regression, not classification

⚡ Acceleration

Massive biomedical data available

3. Problem Definition

Regression Task

Predict binding affinity $pKd = -\log_{10}(Kd)$

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Model: Drug Graph + Protein Embeddings → pKd
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Drug Input

2D Molecular Graph

Atoms as nodes
Bonds as edges

Protein Input

Multi-Modal

3D pocket graphs
Sequence embeddings

4. Existing Works

1. DeepDTA

1D CNNs on SMILES & sequences

Davis: CI 0.882 | **KIBA:** CI 0.892

2. GraphDTA

GNN for drugs, CNN for proteins

Davis: CI 0.876 | **KIBA:** CI 0.888

5. Dataset Descriptions

Davis Dataset

68 Drugs
361 Proteins
24.5K Interactions

Kinase proteins

KIBA Dataset

2.1K Drugs
229 Proteins
117.1K Interactions

Multiple families

6. Contribution & Goal

Research Domain

Computational Drug Discovery

Technical Approach

Graph Neural Networks

Project Goal: Address drug-target affinity prediction problem with better insights through multi-scale graph representations and enhanced interpretability. We aim to slightly improve current metrics while maintaining high interpretability through atom-residue interaction analysis.

7. Project Plan (Part 1: 25% - 50%)

25% – Data Curation & Pipeline Setup

Davis dataset ingestion and protein-ligand mapping
Preprocessing: SMILES to Graph
Protein sequence to ESM embeddings
End-to-end pipeline with sample training

50% – Baseline Implementation & Training

Baseline model fully implemented
Inference pipeline functional
Frontend integration started
Data flow and pipeline setup demonstration

8. Project Plan (Part 2: 75% - 100%)

75% – Model Optimization & Comparison

Cross-modal and joint attention implemented
Ablation studies: 3D structural data vs. 1D sequence embeddings
Model fine-tuning (reaching ≥80% CI)
User interface integration completed

100% – Evaluation & Final Submission

Comprehensive performance analysis (CI, MSE metrics)
Interpretability: atom–residue interactions in kinase pocket
Final presentation, report, and video demonstration

9. Initial Progress

✓ Completed

Dataset found (Davis & KIBA)
Compound process done

⚙ In Progress

Initial graph structure code started
Drug graph construction (2D)
AlphaFold2 protein pocket extraction

~15-20% complete. Foundation laid. Graph construction phase active.